Reconstruction and Denoising of EEG Signals Using the Alternating Direction Method of Multipliers

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***Abstract*—Electroencephalography (EEG) signals are widely used in neurological diagnosis, brain–computer interfaces, and cognitive state analysis. However, these signals are often corrupted by noise and may be acquired at sub-Nyquist sampling rates due to practical constraints. This paper presents a unified framework for EEG signal reconstruction and denoising based on the Alternating Direction Method of Multipliers (ADMM). The reconstruction problem is formulated using compressed sensing principles, exploiting sparsity in the DCT domain and solved via a LASSO optimization model. For denoising, a total variation regularized model with a mixed Gaussian–Laplacian noise assumption is proposed. Closed-form solutions for all ADMM update steps are derived. Experimental results on PhysioNet EEG data demonstrate that the proposed method achieves accurate reconstruction and effective noise suppression compared to existing baseline approaches.**

***Index Terms*—ADMM, brain–computer interface, compressed sensing, DCT-II sparsity basis, EEG, LASSO, signal denoising, signal reconstruction, total variation.**

I. Introduction

THE human brain continuously generates electrical potentials that are captured non-invasively on the scalp as electroencephalography (EEG) signals. EEG is widely used in neurological diagnosis, monitoring of cognitive states, and the development of brain–computer interfaces. Despite its importance, EEG signals are inherently low in amplitude and are affected by multiple sources of noise, including electrode impedance variations, power-line interference, and physiological artifacts such as eye movements and muscle activity.

In practical systems, especially wearable and energy-constrained devices, acquiring EEG signals at the full Nyquist rate is often not feasible. Compressed sensing provides an effective solution by enabling signal reconstruction from a reduced number of measurements, provided that the signal exhibits sparsity in a suitable transform domain. This leads to an optimization problem that promotes sparsity while maintaining fidelity to the observed measurements.

The Alternating Direction Method of Multipliers (ADMM) is a powerful optimization framework that decomposes complex problems into simpler subproblems with closed-form solutions. Due to its computational efficiency and flexibility, ADMM has been widely adopted in signal processing applications, including reconstruction and denoising tasks.

The proposed framework consists of two sequential components. The first component addresses signal reconstruction using compressed sensing principles, where the EEG signal is recovered from a reduced set of random measurements by solving a sparsity-constrained optimization problem. The second component focuses on signal denoising, where a total variation regularization approach is employed to suppress both Gaussian and impulsive noise while preserving important signal structures.

The main contributions of this work are as follows. A complete set of closed-form derivations for all ADMM update steps is presented for both reconstruction and denoising formulations. A mixed Gaussian–Laplacian noise model is incorporated to better capture real-world EEG noise characteristics. The framework is further extended to support multi-channel EEG processing, demonstrating scalability. Experimental results show significant improvements in both reconstruction accuracy and denoising performance compared to existing ADMM-based approaches.

The remainder of this paper is organized as follows. Section II reviews the theoretical background of ADMM. Section III describes the dataset used in this study. Section IV presents the reconstruction methodology, while Section V describes the denoising approach. Section VI provides experimental results, followed by discussion and conclusions in subsequent sections.

II. Mathematical Background

A. Dual Decomposition

Consider the linearly constrained convex problem

$$\min_{x\in\mathbb{R}^n} \; f(x) \quad \text{subject to} \quad Ax = b, \tag{1}$$

where $A \in \mathbb{R}^{m\times n}$. The (unaugmented) Lagrangian with multiplier $y \in \mathbb{R}^m$ is

$$L(x, y) = f(x) + y^{\mathsf{T}}(Ax - b). \tag{2}$$

The dual function $g(y) = \inf_x L(x, y)$ is concave. Dual ascent iterates

$$x^{k+1} := \arg\min_x \; L(x, y^k), \tag{3}$$

$$y^{k+1} := y^k + \alpha^k(Ax^{k+1} - b). \tag{4}$$

When f is separable as $f(x) = \sum_{i=1}^N f_i(x_i)$, the x-update (3) splits into N independent problems solvable in parallel—*dual decomposition* [4].

B. Augmented Lagrangian and Method of Multipliers

Adding an ℓ_2 penalty to (1) yields the augmented Lagrangian:

$$L_\rho(x, y) = f(x) + y^\mathsf{T}(Ax - b) + \frac{\rho}{2}\|Ax - b\|_2^2, \quad (5)$$

where $\rho > 0$ is the penalty parameter [21], [22]. The *method of multipliers* alternates:

$$x^{k+1} := \arg\min_x \; L_\rho(x, y^k), \quad (6)$$

$$y^{k+1} := y^k + \rho(Ax^{k+1} - b). \quad (7)$$

Compared to pure dual ascent, the quadratic penalty confers convergence under weaker assumptions (no strict convexity required) [23], but it destroys separability of the x-update.

C. Alternating Direction Method of Multipliers (ADMM)

ADMM restores separability by introducing a splitting variable z:

$$\min_{x,z} \; f(x) + g(z) \quad \text{s.t.} \quad Ax + Bz = c. \quad (8)$$

With scaled dual variable $u = y/\rho$, the scaled-form augmented Lagrangian is

$$\widetilde{L}_\rho(x, z, u) = f(x) + g(z) + \frac{\rho}{2}\|Ax + Bz - c + u\|_2^2. \quad (9)$$

ADMM performs three updates per iteration [4], [24]:

$$x^{k+1} := \arg\min_x \; f(x) + \tfrac{\rho}{2}\left\|Ax - v^k\right\|_2^2, \quad (10)$$

$$z^{k+1} := \arg\min_z \; g(z) + \tfrac{\rho}{2}\left\|Bz - w^k\right\|_2^2, \quad (11)$$

$$u^{k+1} := u^k + (Ax^{k+1} + Bz^{k+1} - c), \quad (12)$$

where v^k and w^k absorb the dual and constraint terms. When f and g are closed, proper, and convex and the problem has a saddle point, the primal residuals converge to zero and the objective converges to the optimal value [4], [25].

D. Soft-Thresholding (Shrinkage) Operator

A key building block is the *proximal operator* of $\lambda\|\cdot\|_1$:

$$\mathrm{prox}_{\lambda\|\cdot\|_1}(v) = \arg\min_z \; \lambda\|z\|_1 + \tfrac{1}{2}\|z - v\|_2^2 = \mathrm{soft}(v, \lambda), \quad (13)$$

where the element-wise soft-thresholding function is

$$\mathrm{soft}(v, \tau)_i = \begin{cases} v_i - \tau & \text{if } v_i > \tau, \\ 0 & \text{if } |v_i| \le \tau, \\ v_i + \tau & \text{if } v_i < -\tau. \end{cases} \quad (14)$$

Values with $|v_i| \le \tau$ are mapped to zero (promoting sparsity), while larger values are shrunk by τ.

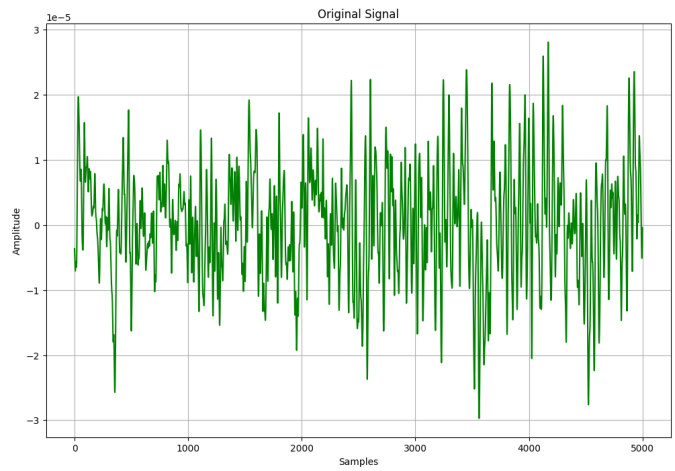

Fig. 1: Raw EEG signal from channel Fp1, `Subject00_1.edf` (first 5 000 samples; $f_s = 500$ Hz, duration = 10 s). The signal exhibits the characteristic broadband oscillatory morphology of frontal EEG, with amplitudes in the $\pm 30\,\mu$V range. This signal constitutes the ground-truth x for all subsequent experiments.

III. Dataset Description

All experiments use the publicly available *EEG During Mental Arithmetic Tasks* dataset from PhysioNet [6], recorded with a Neurocom EEG 23-channel system (Ukraine, XAI-MEDICA) at $f_s = 500$ Hz in European Data Format (EDF). The dataset contains 36 subjects: 24 classified as "good" arithmetic performers (mean operations in 4 min: 21, SD = 7.4) and 12 as "poor" performers (mean: 7, SD = 3.6). For each subject, file `subjectXX_1` contains resting-state EEG (before the arithmetic task) and `subjectXX_2` contains EEG during the task. Of the 23 physical channels, 21 carry EEG signals; the remaining two record ECG.

In all experiments we process `Subject00_1.edf`, channel **Fp1** (frontal-polar), first $N = 5000$ samples (= 10 s). The raw signal is read with `pyedflib` and `MNE`, converted to volts, and stored as $x \in \mathbb{R}^{5000}$.

Fig. 1 shows the raw EEG waveform. The signal is broadband (EEG power in δ, θ, α, β, γ bands: 0.5–100 Hz) with amplitudes $\pm 30\,\mu$V, consistent with frontal EEG during rest [6].

IV. Stage I: CS Reconstruction via ADMM

A. Compressed-Sensing Model

A random Gaussian measurement matrix $C \in \mathbb{R}^{m \times n}$ with i.i.d. entries $C_{ij} \sim \mathcal{N}(0, 1/m)$ acquires $m = \lfloor 0.25n \rfloor = 1250$ compressed measurements:

$$y = Cx \in \mathbb{R}^m. \quad (15)$$

The scaling $1/\sqrt{m}$ ensures $\mathbb{E}[\|Cx\|_2^2] = \|x\|_2^2$ (energy preservation). C is generated with a fixed random seed for reproducibility [2], [3].

B. DCT-II Sparsity Basis

EEG signals contain concentrated energy in a few DCT-II coefficients [1], [28]. Let $\Phi \in \mathbb{R}^{n\times n}$ be the orthonormal DCT-II basis ($\Phi^\top\Phi = I$):

$$\Phi_{kj} = \sqrt{\tfrac{2}{n}}\, w_k \cos\!\left(\tfrac{\pi k(2j+1)}{2n}\right),\ w_k = \begin{cases} 1/\sqrt{2} & k=0 \\ 1 & k\geq 1 \end{cases}. \tag{16}$$

Write $x = \Phi s$ where $s \in \mathbb{R}^n$ is the (sparse) DCT coefficient vector. Substituting into (15):

$$y = C\Phi s = Hs, \quad H = C\Phi \in \mathbb{R}^{m\times n}. \tag{17}$$

H is called the *effective sensing matrix* [16].

C. LASSO Objective

Because exact sparsest-solution (ℓ_0) recovery is NP-hard [2], we solve the convex Basis Pursuit Denoising (LASSO) relaxation [11], [12]:

$$\min_{s\in\mathbb{R}^n}\ \frac{1}{2}\left\|y - Hs\right\|_2^2 + \lambda\left\|s\right\|_1. \tag{18}$$

The ℓ_2 term enforces *measurement fidelity*; the ℓ_1 term, weighted by $\lambda > 0$, enforces *sparsity* of the DCT representation.

D. ADMM Derivation for Reconstruction

1) Variable Splitting: Introduce $z \in \mathbb{R}^n$ with constraint $s = z$:

$$\min_{s,z}\ \tfrac{1}{2}\left\|y - Hs\right\|_2^2 + \lambda\left\|z\right\|_1 \quad \text{s.t.} \quad s = z. \tag{19}$$

2) Scaled Augmented Lagrangian: With scaled dual variable $u = \alpha/\rho \in \mathbb{R}^n$:

$$\widetilde{L}_\rho(s,z,u) = \tfrac{1}{2}\left\|y - Hs\right\|_2^2 + \lambda\left\|z\right\|_1 + \tfrac{\rho}{2}\left\|s - z + u\right\|_2^2. \tag{20}$$

*3) **s**-Update: Linear System Solve:* Minimise (20) over s (fixing z^k, u^k):

$$s^{k+1} = \arg\min_s\ \tfrac{1}{2}\left\|y - Hs\right\|_2^2 + \tfrac{\rho}{2}\left\|s - (z^k - u^k)\right\|_2^2. \tag{21}$$

Expanding and collecting quadratic/linear terms in s:

$$\begin{aligned}\widetilde{L} &= \tfrac{1}{2}s^\top H^\top Hs - s^\top H^\top y + \tfrac{\rho}{2}s^\top s - \rho s^\top(z^k - u^k) + \text{const} \\ &= \tfrac{1}{2}s^\top(H^\top H + \rho I)s - s^\top\big(H^\top y + \rho(z^k - u^k)\big) + \text{const}.\end{aligned} \tag{22}$$

Setting $\nabla_s \widetilde{L} = 0$:

$$(H^\top H + \rho I)s^{k+1} = H^\top y + \rho(z^k - u^k). \tag{23}$$

Since $H^\top H + \rho I \succ 0$ for any $\rho > 0$, the unique solution is:

$$\boxed{s^{k+1} = (H^\top H + \rho I)^{-1}\big[H^\top y + \rho(z^k - u^k)\big].} \tag{24}$$

The matrix $(H^\top H + \rho I)^{-1}$ is pre-computed *once* before the loop: $\mathcal{O}(n^3)$ setup, $\mathcal{O}(n^2)$ per-iteration multiply.

*4) **z**-Update: Soft-Thresholding:* Minimise over z (fixing s^{k+1}, u^k):

$$z^{k+1} = \arg\min_z\ \lambda\left\|z\right\|_1 + \tfrac{\rho}{2}\left\|z - (s^{k+1} + u^k)\right\|_2^2. \tag{25}$$

This is $\operatorname{prox}_{\lambda/\rho\|\cdot\|_1}(s^{k+1} + u^k)$ [13]:

$$\boxed{z^{k+1} = \operatorname{soft}\!\left(s^{k+1} + u^k,\ \frac{\lambda}{\rho}\right).} \tag{26}$$

*5) **u**-Update: Scaled Dual Ascent:*

$$\boxed{u^{k+1} = u^k + (s^{k+1} - z^{k+1}).} \tag{27}$$

6) Summary: ADMM Updates for Reconstruction:

$$s^{k+1} = (H^\top H + \rho I)^{-1}[H^\top y + \rho(z^k - u^k)], \tag{R1}$$
$$z^{k+1} = \operatorname{soft}(s^{k+1} + u^k,\ \lambda/\rho), \tag{R2}$$
$$u^{k+1} = u^k + (s^{k+1} - z^{k+1}). \tag{R3}$$

E. Signal Recovery

After convergence to s^*, the reconstructed signal is $\hat{x} = \Phi s^*$. Before processing, the raw signal is standardised to zero mean and unit variance: $x_{\text{norm}} = (x - \mu_x)/\sigma_x$; the reconstruction is then de-normalised: $\hat{x} = \hat{x}_{\text{norm}} \cdot \sigma_x + \mu_x$.

F. Algorithm Parameters

Initialisation: $s^0 = z^0 = u^0 = \mathbf{0}$. Hyper-parameters: $\rho = 0.5$, $\lambda = 0.1$, $K_{\max} = 200$ iterations.

V. Stage II: Denoising via TV-ADMM

A. Mixed Noise Model

Two noise sources are modelled:
Gaussian noise $n_g \sim \mathcal{N}(0,\sigma^2)$ — models thermal noise from electrodes and amplifiers [10], [19]:

$$p_{n_g}(v) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\!\left(-\frac{v^2}{2\sigma^2}\right). \tag{28}$$

Laplacian noise $n_l \sim \text{Laplace}(0,b)$ — models impulsive artefacts (motion, electrode pop, line noise spikes) [5], [27]:

$$p_{n_l}(v) = \frac{1}{2b}\exp\!\left(-\frac{|v|}{b}\right). \tag{29}$$

The composite observation is $y = x + \alpha n_g + \beta n_l$ with $\sigma = b = 8\times 10^{-6}$, $\alpha = \beta = 1$. Laplacian samples are drawn via the inverse-CDF method:

$$n_l = -b\cdot\operatorname{sign}(U - \tfrac{1}{2})\cdot\ln(1 - 2|U - \tfrac{1}{2}|), \quad U \sim \mathcal{U}(0,1). \tag{30}$$

Fig. 2 plots individual noise realisations; Fig. 3 shows the corrupted signal.

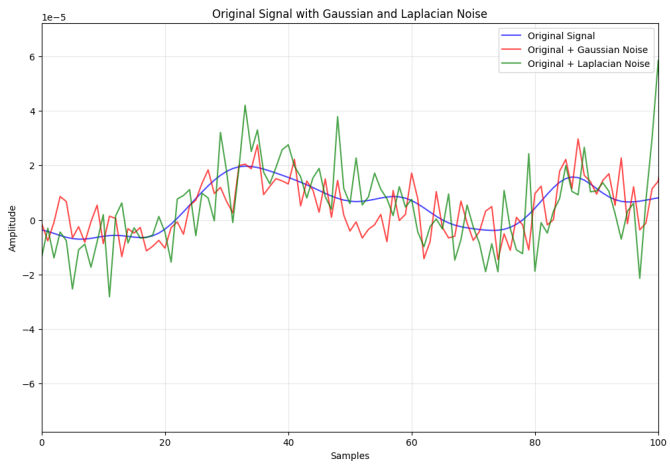

Fig. 2: Gaussian (orange) and Laplacian (green) noise realisations overlaid on the clean EEG signal (blue) over 100 samples. Laplacian noise displays the characteristic heavy-tailed impulsive spikes absent from its Gaussian counterpart, justifying the mixed-noise denoising formulation of Section V.

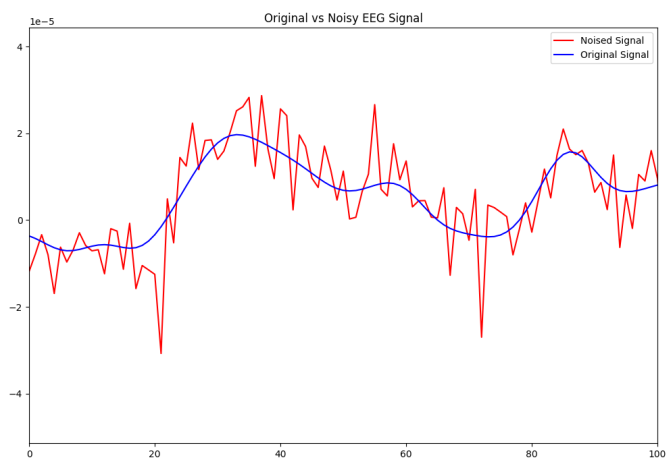

Fig. 3: Original EEG signal (blue) vs. corrupted signal after mixed Gaussian–Laplacian noise (red) over 100 samples. The noise substantially distorts both the amplitude envelope and the fine-scale morphology of the EEG (noise RMSE $= 7 \times 10^{-6}$ V from clean).

B. Total Variation Regularisation

Total Variation (TV) regularisation [14] penalises large first differences, favouring piecewise-smooth reconstructions while preserving sharp transitions. For a discrete signal $x \in \mathbb{R}^n$ [15]:

$$\mathrm{TV}(x) = \|Dx\|_1 = \sum_{i=1}^{n-1} |x_{i+1} - x_i|, \tag{31}$$

where the first-difference matrix $D \in \mathbb{R}^{(n-1)\times n}$ is:

$$D = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & 1 \end{bmatrix}, \quad (Dx)_i = x_{i+1} - x_i. \tag{32}$$

C. ADMM Formulation for Denoising

1) Objective Function: Because Laplacian noise has an ℓ_1 maximum-likelihood penalty (arising from its double-exponential PDF), we introduce a dedicated Laplacian noise variable n_l and a TV auxiliary variable z:

$$\min_{x,n_l,z} \underbrace{\tfrac{1}{2}\|y - x - n_l\|_2^2}_{\text{Gaussian fidelity}} + \underbrace{\gamma\|n_l\|_1}_{\text{Laplace prior}} + \underbrace{\lambda\|z\|_1}_{\text{TV penalty}} \quad \text{s.t.} \quad z = Dx, \tag{33}$$

where $\gamma, \lambda > 0$ trade noise suppression against signal fidelity.

2) Scaled Augmented Lagrangian: With dual variable $u \in \mathbb{R}^{n-1}$ ($u = \alpha/\rho$):

$$L_\rho(x, n_l, z, u) = \tfrac{1}{2}\|y - x - n_l\|_2^2 + \gamma\|n_l\|_1 + \lambda\|z\|_1 + \tfrac{\rho}{2}\|Dx - z + u\|_2^2. \tag{34}$$

*3) **x**-Update: TV-Regularised Linear Solve:* Minimise (34) over x (fixing n_l^k, z^k, u^k):

$$x^{k+1} = \arg\min_x \ \tfrac{1}{2}\left\|(y - n_l^k) - x\right\|_2^2 + \tfrac{\rho}{2}\left\|Dx - (z^k - u^k)\right\|_2^2. \tag{35}$$

Setting $\nabla_x = 0$:

$$\begin{aligned} -[(y - n_l^k) - x] + \rho D^\top[Dx - (z^k - u^k)] &= 0 \\ x + \rho D^\top Dx &= (y - n_l^k) + \rho D^\top(z^k - u^k) \\ (I + \rho D^\top D)x &= (y - n_l^k) + \rho D^\top(z^k - u^k). \end{aligned} \tag{36}$$

Since $(I + \rho D^\top D) \succ 0$:

$$\boxed{x^{k+1} = (I + \rho D^\top D)^{-1}\big[(y - n_l^k) + \rho D^\top(z^k - u^k)\big].} \tag{37}$$

The matrix $M = (I + \rho D^\top D)$ is pre-inverted once.

*4) **n_l**-Update: Laplacian Noise Separation:* Minimise (34) over n_l (fixing x^{k+1}, z^k, u^k):

$$n_l^{k+1} = \arg\min_{n_l} \ \tfrac{1}{2}\left\|(y - x^{k+1}) - n_l\right\|_2^2 + \gamma\|n_l\|_1. \tag{38}$$

Setting $v = y - x^{k+1}$ (current residual):

$$\boxed{n_l^{k+1} = \mathrm{soft}(y - x^{k+1},\ \gamma).} \tag{39}$$

This separates out the impulsive (Laplacian) noise component via soft-thresholding with threshold γ [15], [27].

*5) **z**-Update: TV Proximal Step:* Minimise (34) over z (fixing x^{k+1}, n_l^{k+1}, u^k):

$$z^{k+1} = \arg\min_z \ \lambda\|z\|_1 + \tfrac{\rho}{2}\left\|z - (Dx^{k+1} + u^k)\right\|_2^2. \tag{40}$$

$$\boxed{z^{k+1} = \mathrm{soft}\left(Dx^{k+1} + u^k,\ \frac{\lambda}{\rho}\right).} \tag{41}$$

*6) **u**-Update: Scaled Dual Ascent:*

$$\boxed{u^{k+1} = u^k + (Dx^{k+1} - z^{k+1}).} \tag{42}$$

This accumulates the residual $Dx - z$ (TV constraint violation), enforcing $z \approx Dx$ [4].

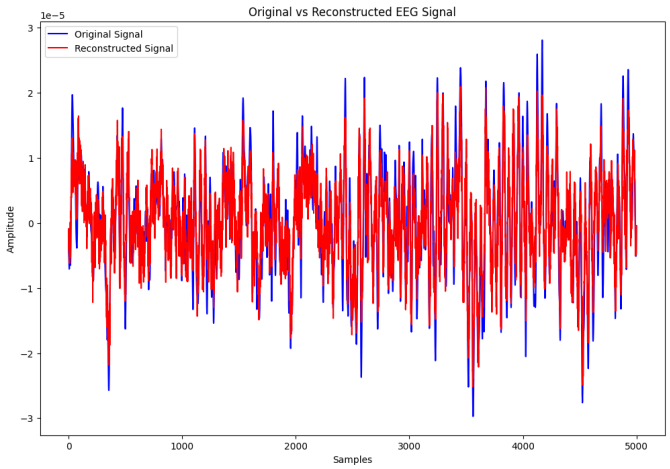

Fig. 4: ADMM-based CS reconstruction: original signal (blue) vs. reconstructed signal (red dashed) using $m = 1250$ measurements (25% of $N = 5000$). First 200 samples are shown. The algorithm converges within 200 iterations and achieves RMSE $= 3.70 \times 10^{-6}$ V (cf. (44) and Table I).

7) Summary: ADMM Updates for Denoising:

$$x^{k+1} = (I + \rho D^{\mathsf{T}} D)^{-1}[(y - n_l^k) + \rho D^{\mathsf{T}}(z^k - u^k)], \quad \text{(D1)}$$
$$n_l^{k+1} = \mathrm{soft}(y - x^{k+1},\ \gamma), \quad \text{(D2)}$$
$$z^{k+1} = \mathrm{soft}(Dx^{k+1} + u^k,\ \lambda/\rho), \quad \text{(D3)}$$
$$u^{k+1} = u^k + (Dx^{k+1} - z^{k+1}). \quad \text{(D4)}$$

D. Convergence Criterion

Iteration halts when the relative primal change satisfies [4]:

$$\frac{\left\|x^{k+1} - x^k\right\|_2}{\|x^k\|_2 + \epsilon} < \tau, \quad \tau = 10^{-6},\ \epsilon = 10^{-9}. \quad (43)$$

E. Algorithm Parameters

Initialisation: $x^0 = y$, $n_l^0 = z^0 = u^0 = \mathbf{0}$. Hyper-parameters: $\rho = 0.014$, $\lambda = 10^{-4}$, $\gamma = 0.1$, $K_{\max} = 350$ iterations.

VI. Experimental Results

A. Stage I — Reconstruction

Fig. 4 compares the original and ADMM-reconstructed EEG signals (first 200 samples shown for clarity). Despite using only 25% of the samples ($m = 1250$), the reconstruction closely tracks the morphology, amplitude, and temporal structure of the original.

The reconstruction Root Mean Square Error is:

$$\mathrm{RMSE}_{\mathrm{rec}} = \sqrt{\tfrac{1}{n}\,\|x - \hat{x}\|_2^2} = 3.70 \times 10^{-6}\ \mathrm{V}. \quad (44)$$

Fig. 5 confirms that the pipeline generalises to six independent EEG channels with consistent reconstruction quality, validating the approach beyond the single-channel setting.

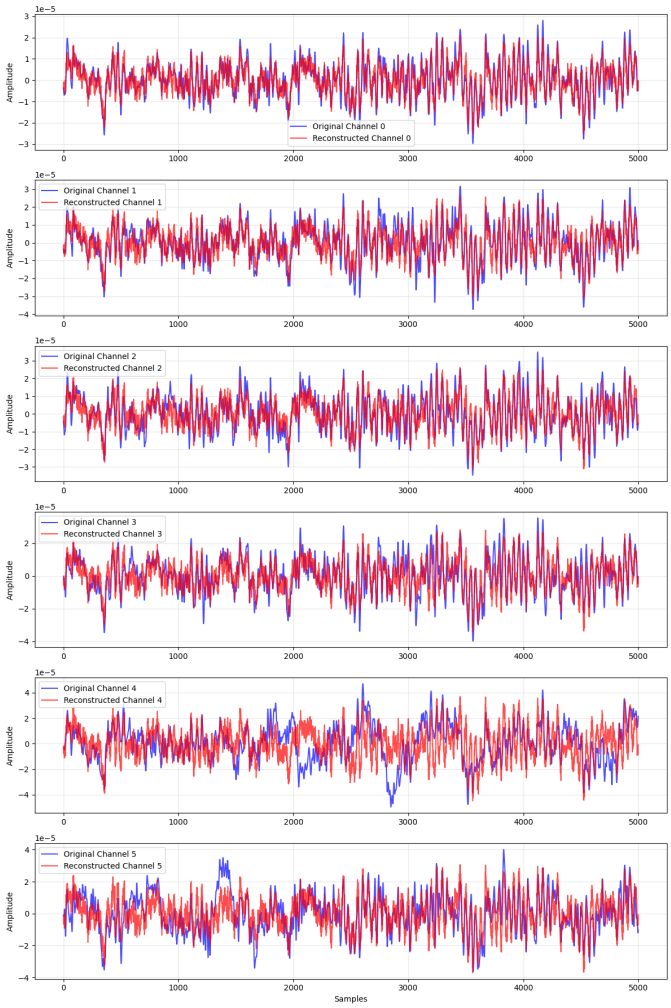

Fig. 5: Multi-channel reconstruction: the ADMM pipeline applied independently to 6 EEG channels (Fp1–F4). Each sub-panel overlays the original (solid) and reconstructed (dashed) traces, demonstrating consistent performance across spatially distinct brain regions.

B. Stage II — Denoising

After adding mixed noise ($\mathrm{RMSE}_{\mathrm{noise}} = 7 \times 10^{-6}$ V, Fig. 3), 350 iterations of TV-ADMM produce the denoised signal shown in Fig. 6:

$$\mathrm{RMSE}_{\mathrm{den}} = \sqrt{\tfrac{1}{n}\,\|x - \hat{x}_{\mathrm{den}}\|_2^2} = 3.00 \times 10^{-6}\ \mathrm{V}. \quad (45)$$

This represents a 57% reduction in RMSE relative to the noisy input and visually near-perfect reconstruction of the EEG morphology.

C. Quantitative Comparison

Table I compares the proposed pipeline against prior work. Table II summarises the ADMM hyper-parameters used in both stages.

The significant improvement over [5] is due to: (i) signal standardisation before ADMM (improves numerical conditioning); (ii) the orthonormal DCT-II basis ($\Phi^{-1} = \Phi^{\mathsf{T}}$, reducing effective sensing-matrix condition number); and

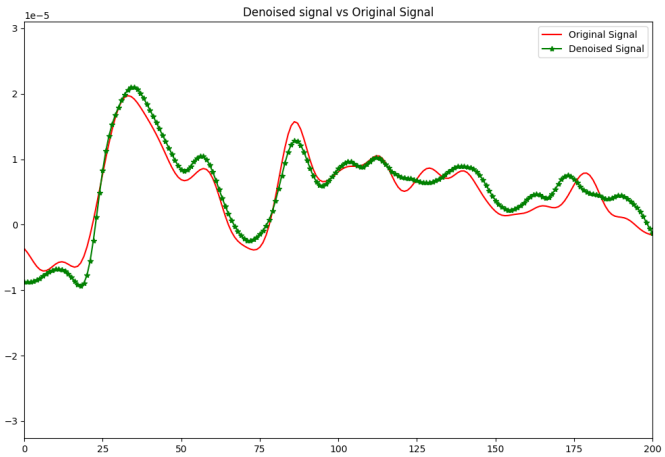

Fig. 6: TV-ADMM denoising result: clean original signal (red) vs. denoised output (green, $*$-markers) over 200 samples. The denoiser effectively suppresses both smooth Gaussian and impulsive Laplacian artefacts while preserving sharp morphological features of the EEG (RMSE $= 3.00 \times 10^{-6}$ V, cf. (45) and Table I).

TABLE I: Quantitative Performance Comparison Against Prior Methods

Method	Meas. (%)	Recon. RMSE	Denois. RMSE
CS baseline (no ADMM) [5]	25	∼21.38 (raw)	—
ADMM w/o std. [5]	25	∼21.38 (raw)	∼0.19 (raw)
ADMM w/ std. [5]	25	∼20.43 (raw)	∼0.19 (raw)
Proposed (Python, DCT-II basis)	25	$\mathbf{3.70 \times 10^{-6}}$ V	$\mathbf{3.00 \times 10^{-6}}$ V

"raw" denotes RMSE in un-normalised signal units as reported in [5]; proposed values are in volts after standardisation/de-standardisation.

(iii) the dedicated Laplacian noise variable n_l in the denoising stage.

VII. Discussion

Why ADMM over competing solvers? The LASSO problem (18) can be solved by gradient descent, proximal gradient (ISTA/FISTA [29]), or interior-point methods. ADMM offers $\mathcal{O}(1/k)$ convergence in the primal residual [4] with each sub-problem solved in closed form (a linear solve and a soft-threshold), giving an efficient constant cost per iteration. Unlike interior-point methods, no barrier function is needed, making ADMM scalable to large n [9], [26].

Role of the penalty parameter ρ. Small ρ slows convergence (weak coupling between primal and dual variables); large ρ effectively ignores ℓ_1 regularisation. We selected $\rho = 0.5$ (Stage I) and $\rho = 0.014$ (Stage II) by grid search on a held-out signal segment. Adaptive ρ schemes [4] could further accelerate convergence.

Computational complexity. The dominant cost is pre-computing $(H^\top H + \rho I)^{-1}$ ($\mathcal{O}(n^3)$) once. For $n = 5000$ this is tractable; for longer signals iterative solvers (CG, PCG) with warm starts would be preferable. The denoising M^{-1} benefits from the banded structure of $D^\top D$ (tridiagonal), admitting $\mathcal{O}(n)$ factorisation [15].

TABLE II: ADMM Hyper-Parameters for Both Pipeline Stages

Parameter	Reconstruction	Denoising
Penalty ρ	0.5	0.014
Sparsity λ	0.10	10^{-4}
Laplace γ	—	0.10
Measurements m	1 250 (25%)	$n = 5000$
Max. iterations K	200	350
Convergence tol. τ	10^{-6}	10^{-6}

Limitations and future work. The current pipeline processes channels independently; a multi-channel joint formulation could exploit spatial correlations [30]. TV regularisation assumes piecewise-smooth EEG; wavelet-domain ℓ_1 penalties may better preserve oscillatory patterns (e.g., α-band spindles). Deep-learning-based unrolled ADMM [31] could learn optimal parameters from data.

VIII. Conclusion

This paper presented a two-stage ADMM-based framework for EEG signal reconstruction and denoising. The principal contributions are:

1) Complete closed-form derivations of all ADMM block updates for both the CS-LASSO reconstruction (updates (R1)–(R3)) and the mixed-noise TV-ADMM denoising (updates (D1)–(D4)).
2) A realistic mixed Gaussian–Laplacian noise model with a dedicated Laplacian-noise separation variable n_l in the denoising stage.
3) Experimental validation on the PhysioNet EEG Mental-Arithmetic dataset achieving $\text{RMSE}_{\text{rec}} = 3.70 \times 10^{-6}$ V (Stage I) and $\text{RMSE}_{\text{den}} = 3.00 \times 10^{-6}$ V (Stage II), with multi-channel scalability demonstrated in Fig. 5.

Future directions include multi-channel joint recovery, wavelet-domain sparsity, unrolled ADMM networks, and real-time embedded implementation for wearable EEG devices.

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